Today's Agenda

- **Spatial image filtering**
  - Linear filters
    - –Image Smoothing
    - –Image sharpening
  - Nonlinear filter
- **Fourier transform**

Image Sharpening by Unsharp Masking and Highboost Filtering

1. **Blur the original image**
2. **Subtract the blurred image from the original to get the mask**
3. **Add the mask to the original**

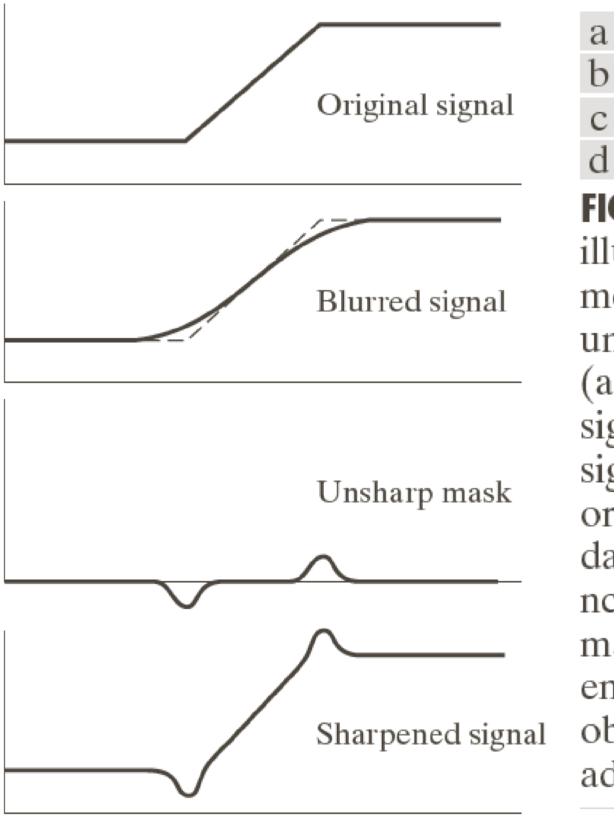


FIGURE 3.39 1-D illustration of the mechanics of unsharp masking. (a) Original signal. (b) Blurred signal with original shown dashed for reference. (c) Unsharp mask. (d) Sharpened signal, obtained by adding (c) to (a).

Image Sharpening by Unsharp Masking and Highboost Filtering

$$g(x,y) = f(x,y) + k*\left(f(x,y) - \bar{f}(x,y)\right) \quad k \geq 0$$

When $k > 1$, it becomes a highboost filtering.

Example: when $k = 1$

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} + \left(\begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} - \frac{1}{9} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \right) = \frac{1}{9} \begin{bmatrix} -1 & -1 & -1 \\ -1 & 17 & -1 \\ -1 & -1 & -1 \end{bmatrix}$$

Sum of the coefficients is 1

An Example

Original

Blurred

Unsharp mask

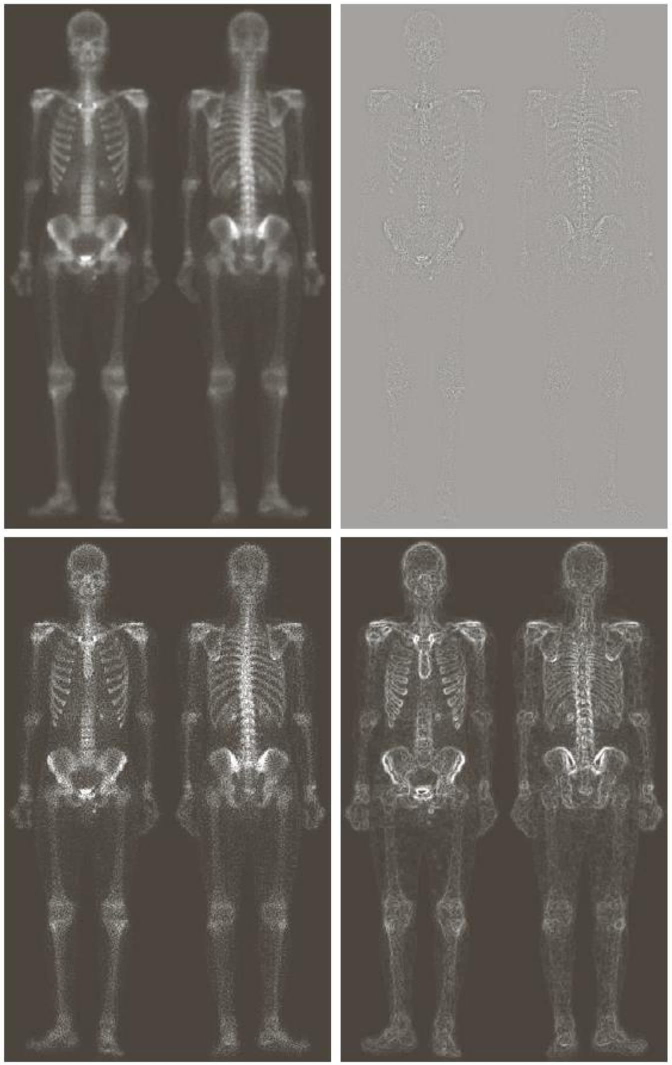
Unsharp masking k=1

Highboost filitering k=4.5

a
b
c
d
e

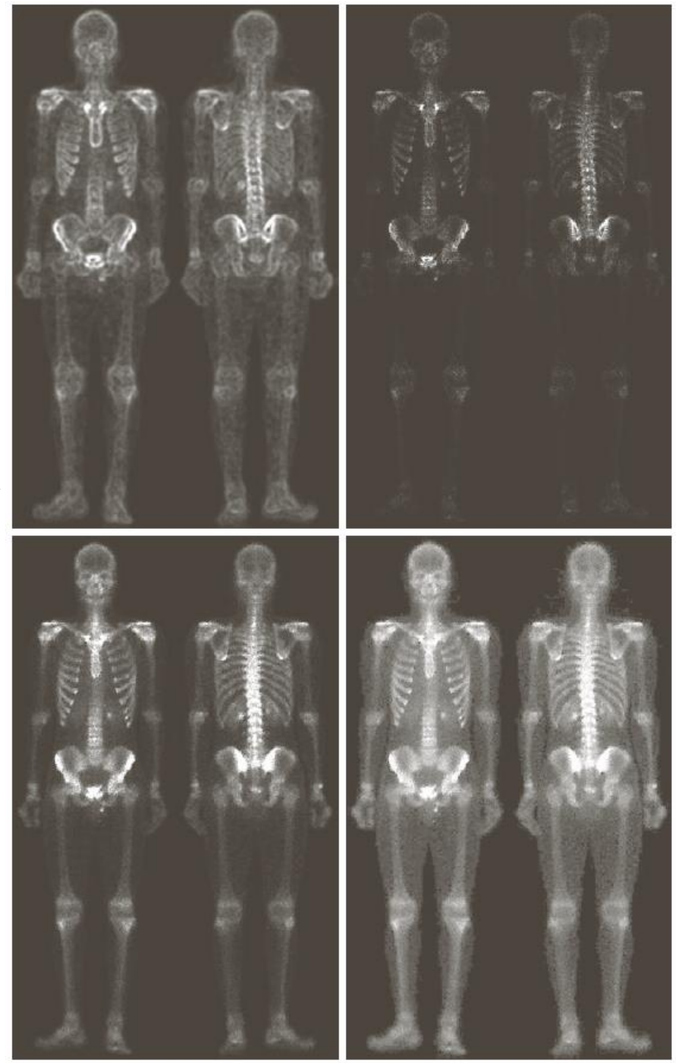
FIGURE 3.40
(a) Original image.
(b) Result of blurring with a Gaussian filter.
(c) Unsharp mask. (d) Result of using unsharp masking.
(e) Result of using highboost filtering.

Combining Spatial Enhancement Methods



a b
c d

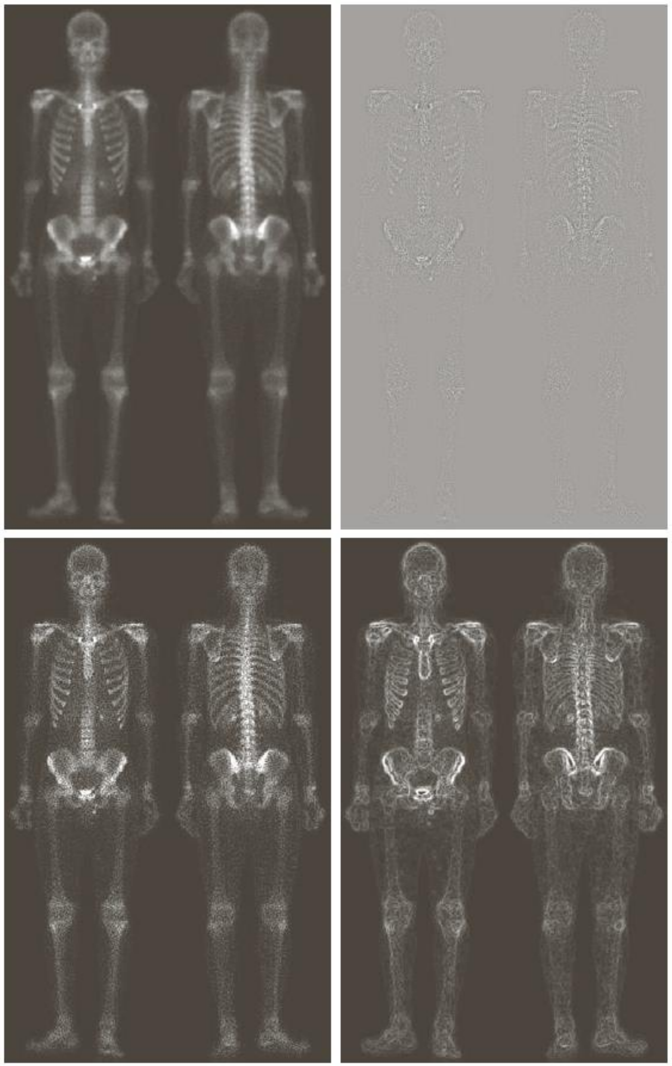
FIGURE 3.43
(a) Image of whole body bone scan.
(b) Laplacian of (a). (c) Sharpened image obtained by adding (a) and (b).
(d) Sobel gradient of (a).



e f
g h

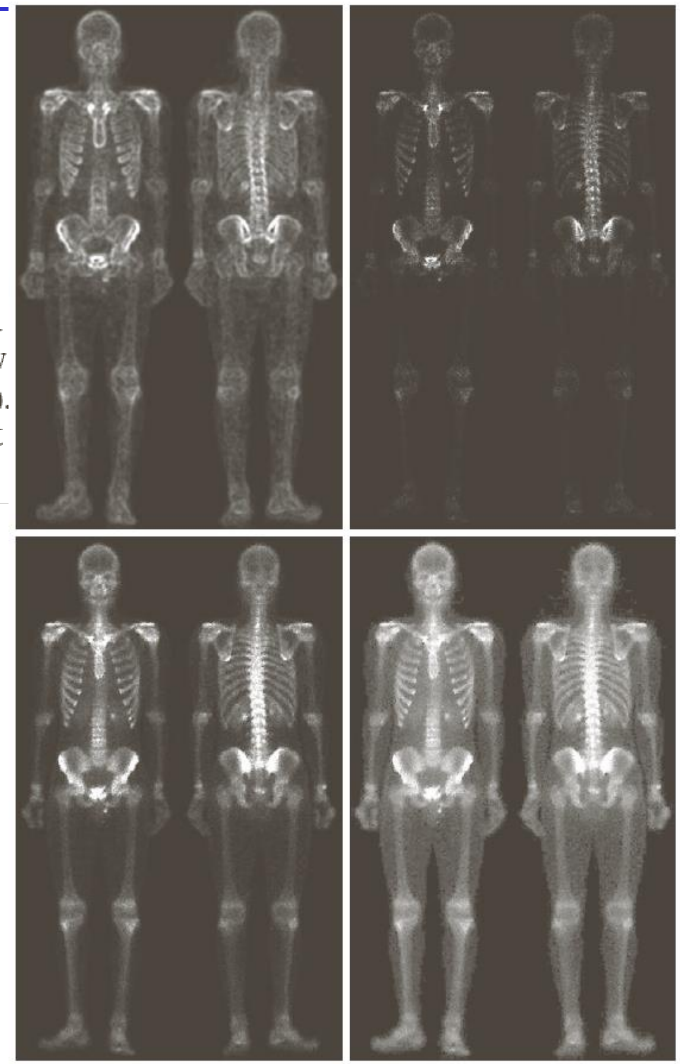
FIGURE 3.43
(Continued)
(e) Sobel image smoothed with a 5 × 5 averaging filter. (f) Mask image formed by the product of (c) and (e).
(g) Sharpened image obtained by the sum of (a) and (f). (h) Final result obtained by applying a power-law transformation to (g). Compare (g) and (h) with (a). (Original image courtesy of G.E. Medical Systems.)

Combining Spatial Enhancement Methods



a b
c d

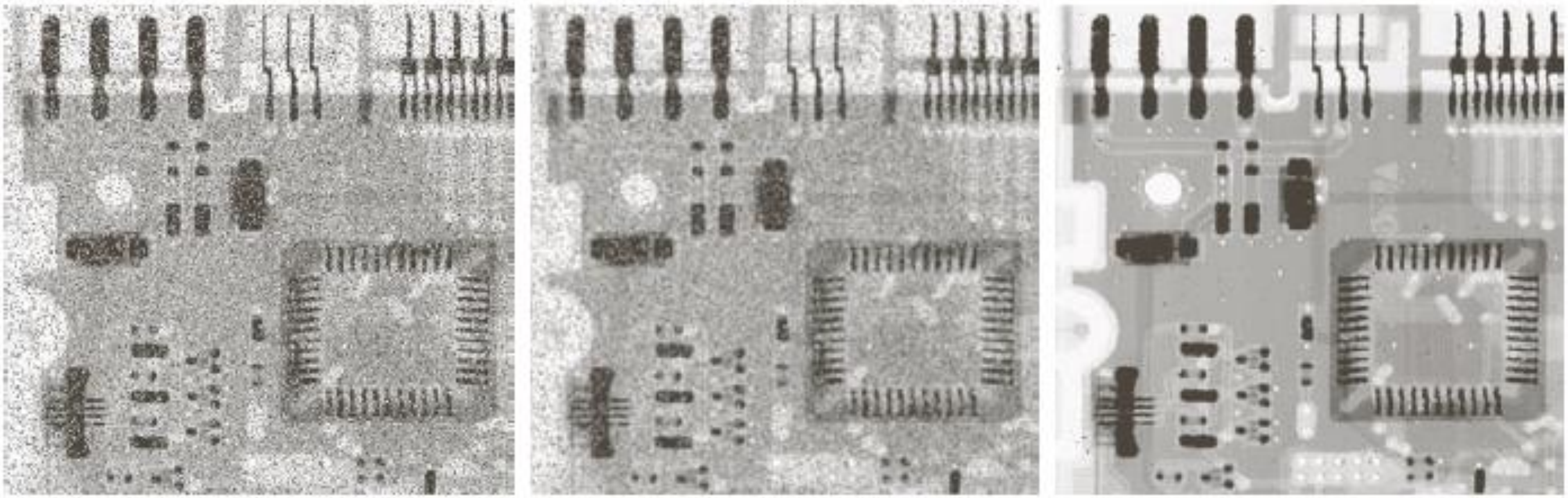
FIGURE 3.43
(a) Image of whole body bone scan. (b) Laplacian of (a). (c) Sharpened image obtained by adding (a) and (b). (d) Sobel gradient of (a).



e f
g h

FIGURE 3.43
(Continued)
(e) Sobel image smoothed with a 5×5 averaging filter. (f) Mask image formed by the product of (c) and (e). (g) Sharpened image obtained by the sum of (a) and (f). (h) Final result obtained by applying a power-law transformation to (g). Compare (g) and (h) with (a). (Original image courtesy of G.E. Medical Systems.)

Order-Statistic (Nonlinear) Filtering



a b c

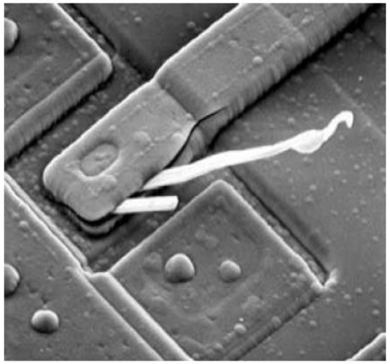
FIGURE 3.35 (a) X-ray image of circuit board corrupted by salt-and-pepper noise. (b) Noise reduction with a 3 × 3 averaging mask. (c) Noise reduction with a 3 × 3 median filter. (Original image courtesy of Mr. Joseph E. Pascente, Lixi, Inc.)

Order-statistic filtering – rank the pixel values in the filter window and assign the center pixel according to the property of the filter

- Median
- Min/max

Why We Need Fourier Transform

- **Filtering in frequency domain**



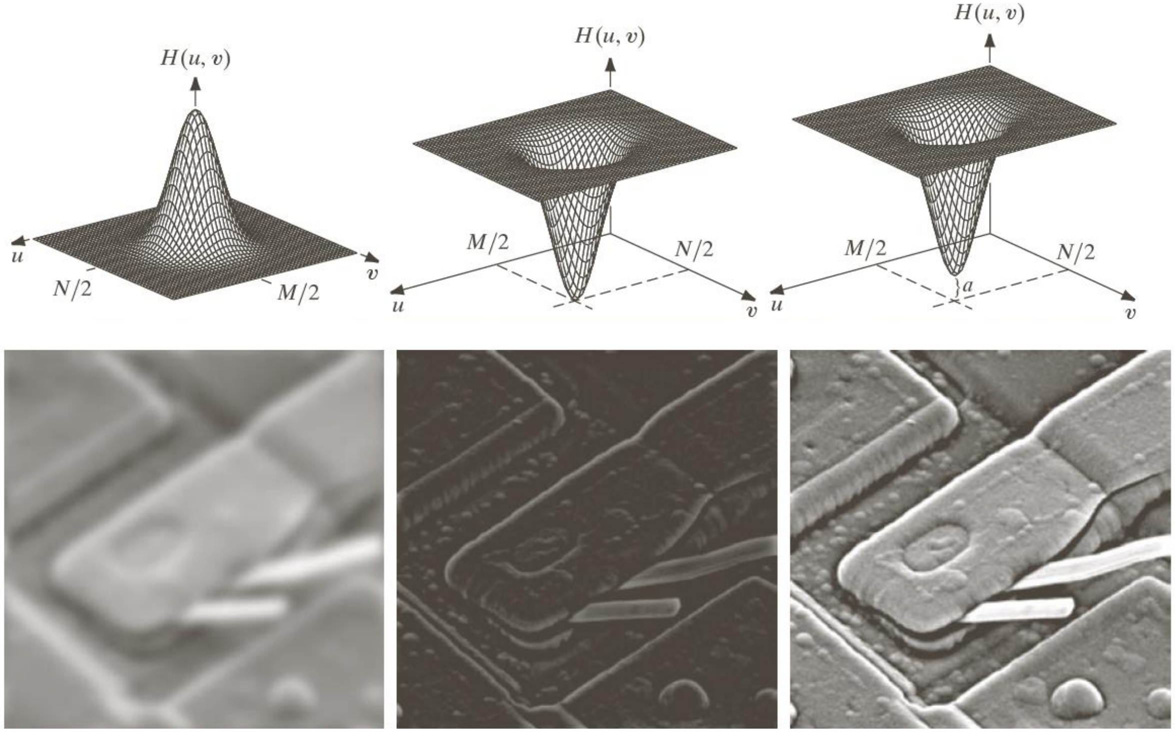
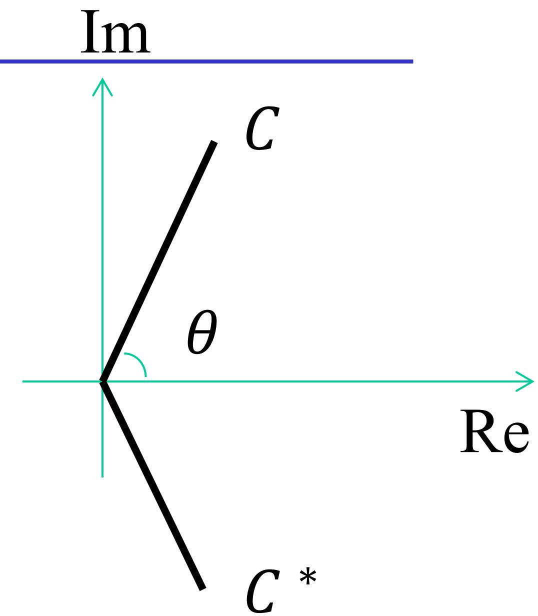


Image smoothing　　Edge　　Image sharpening

- **Efficient computation for convolution**

Preliminary Concepts



Complex number $C = R + jI \quad j = \sqrt{-1}$

Conjugate $C^* = R - jI$

Polar coordinate representation

$$C = |C|(\cos\theta + j\sin\theta)$$

$$|C| = \sqrt{R^2 + I^2}, \qquad \theta = \arctan(I/R)$$

Euler's formula $e^{j\theta} = \cos\theta + j\sin\theta \qquad C = |C|e^{j\theta}$

Concept of Fourier Series And Transforms

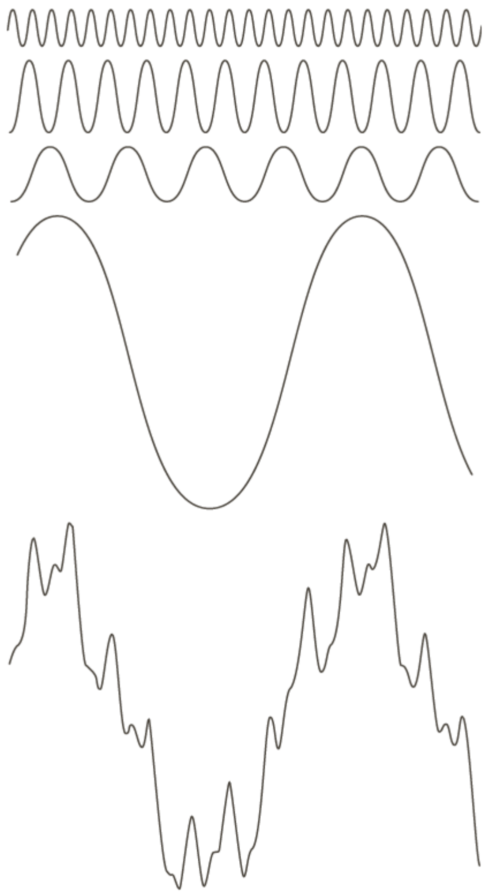


FIGURE 4.1 The function at the bottom is the sum of the four functions above it. Fourier's idea in 1807 that periodic functions could be represented as a weighted sum of sines and cosines was met with skepticism.

Fourier series: any **periodic** function can be represented by a **discrete** weighted sum of sines and cosines

Concept of Fourier Series And Transforms

Fourier series: any periodic function can be represented by a discrete weighted sum of sines and cosines

Fourier transform: an arbitrary function with **finite duration** (**non-periodic** function) can be expressed by a weighted **integrals** of sines and cosines

Fourier transform is more general!

Fourier Series

$f(t)$ **is a continuous function with period** T**, we have**

$$f(t) = \sum_{n=-\infty}^{+\infty} c_n e^{\frac{j2\pi nt}{T}} \Rightarrow \cos\frac{2\pi n}{T}t + j\sin\frac{2\pi n}{T}t$$

Coefficient

Discrete frequency

Where

$$c_n = \frac{1}{T}\int_{-T/2}^{T/2} f(t)\, e^{-\frac{j2\pi nt}{T}}\, dt, \quad n = 0, \pm 1, \pm 2, \ldots$$

Video demo

https://en.wikipedia.org/wiki/Fourier_transform#/media/File:Fourier_transform_time_and_frequency_domains_(small).gif

Fourier Transform in 1D

$f(t)$ is an arbitrary non-periodic function and can be represented by

$$f(t) = \int_{-\infty}^{\infty} F(\mu) e^{j2\pi\mu t} d\mu$$

Coefficient

Continuous frequency

where

$$F(\mu) = \int_{-\infty}^{\infty} f(t) e^{-j2\pi\mu t} dt$$

Fourier Transform in 1D

$f(t)$ **is an arbitrary non-periodic function and can be represented by**

$$f(t) = \int_{-\infty}^{\infty} F(\mu) e^{j2\pi\mu t} d\mu$$

Coefficient

Continuous frequency

where

$$F(\mu) = \int_{-\infty}^{\infty} f(t) e^{-j2\pi\mu t} dt$$

Fourier series

Discrete frequency

$$f(t) = \sum_{n=-\infty}^{+\infty} c_n e^{\frac{j2\pi nt}{T}}$$

$$c_n = \frac{1}{T} \int_{-T/2}^{T/2} f(t) e^{-\frac{j2\pi nt}{T}} dt$$

Fourier Transform in 1D

Spatial domain → Frequency domain

$$F(\mu) = \int_{-\infty}^{\infty} f(t) e^{-j2\pi\mu t} dt$$ Forward transform

Frequency domain → Spatial domain

$$f(t) = \int_{-\infty}^{\infty} F(\mu) e^{j2\pi\mu t} d\mu$$ Inverse transform

Fourier transform pair

Basic Properties of FT

Linearity $h(t) = af(t) + bg(t) \leftrightarrow H(\mu) = aF(\mu) + bG(\mu)$

Translation $h(t) = f(t - t_0) \leftrightarrow H(\mu) = e^{-j2\pi t_0 \mu} F(\mu)$

Translation in spatial domain → Rotation in frequency domain

Modulation $h(t) = e^{j2\pi\mu_0 t} f(t) \leftrightarrow H(\mu) = F(\mu - \mu_0)$

Rotation in spatial domain → Translation in frequency domain

Basic Properties of FT

Scaling

$$h(t) = f(at) \leftrightarrow H(\mu) = \frac{1}{|a|} F(\frac{\mu}{a})$$

Conjugation

$$h(t) = f^*(t) \leftrightarrow H(\mu) = F^*(-\mu)$$

Symmetry

$$f(t) \leftrightarrow F(\mu) \Rightarrow F(t) \leftrightarrow f(-\mu)$$

FT of Simple Functions

$$f(t) = \begin{cases} A & -\dfrac{w}{2} \leq t \leq \dfrac{w}{2} \\ 0 & otherwise \end{cases}$$

$$F(\mu) = \frac{A}{\pi\mu} \sin \pi w \mu = Aw \frac{\sin \pi w \mu}{\pi w \mu} = Aw\ sinc(\pi w \mu)$$

FT of a Rectangle Function

Rectangle function → Sinc function

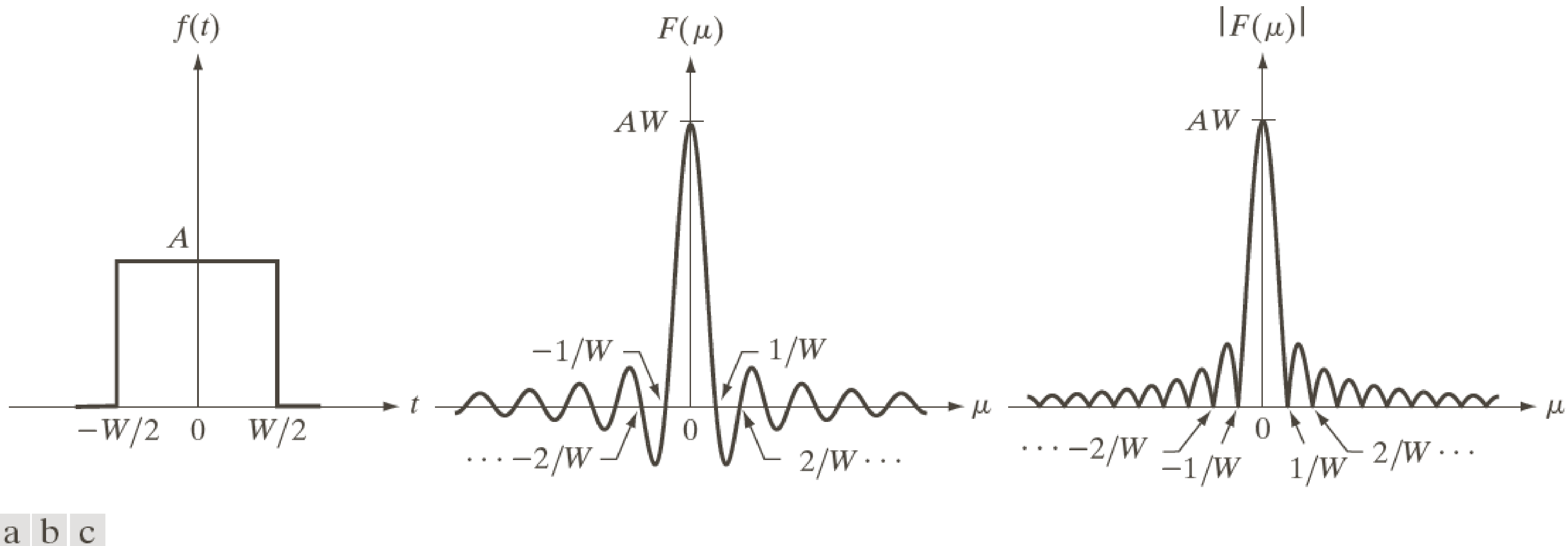


FIGURE 4.4 (a) A simple function; (b) its Fourier transform; and (c) the spectrum. All functions extend to infinity in both directions.